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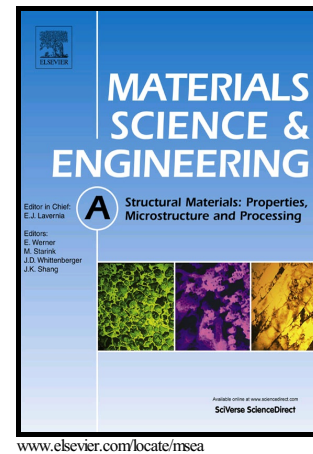
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Deformation behavior of CP-titanium: Experiment and Crystal plasticity modeling

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Abstract

The deformation behavior of commercially pure titanium is studied using experiments and a crystal plasticity model. Compression tests along the rolling, transverse, and normal-directions, and tensile tests along the rolling and transverse directions are performed at room temperature to study the activation of slip and twinning in the hexagonal closed packed titanium. A detailed EBSD based statistical analysis of the microstructure is performed to develop statistics of both {10-12} tensile and {11-22} compression twins. A simple Monte Carlo (MC) twin variant selection criterion is proposed within the framework of the viscoplastic self-consistent (VPSC) model with a dislocation density (DD) based law used to describe dislocation hardening. In the model, plasticity is accommodated by prismatic, basal and pyramidal slip modes, and {10-12} tensile and {11-22} compression twinning modes. The VPSC-MC model successfully captures the experimentally observed activation of low Schmid factor twin variants for both tensile and compression twins modes. The model also predicts macroscopic stress-strain response, texture evolution and twin volume fraction that are in agreement with experimental observations.

Keywords: crystal plasticity, twin variant selection, titanium, dislocation density

1. Introduction

Titanium and its alloys are used in several industries, including automotive and aerospace, due to their high specific strength. A lack of available crystallographic slip systems in hexagonal close packed (HCP) titanium necessitates the activation of twinning to accommodate any arbitrary deformation [1-3]. In HCP titanium, {10-12} twins normally accommodate extension along the c-axis and are frequently referred to as tensile twins, and

{11-22} twins normally accommodate contraction along the c-axis and are frequently referred to as compression twins. While dislocation slip is rather homogeneous over the crystals and involves gradual lattice reorientation, deformation twins are localized, lamellar shaped domains within the crystal resulting in a significant lattice reorientation [3-5]. For instance, in titanium with the c/a ratio of 1.588, the lattice rotation and local shear strain associated with {10-12} <0111> tensile twins is approximately 85° and 17.4%, respectively. The heterogeneous twinning microstructure greatly affects the polycrystal deformation behavior through: i) reorientation to a harder or softer orientation resulting in the activation of different deformation modes, ii) a Hall-Petch effect due to twin induced grain subdivision, iii) creation of immobile dislocation content within twins due to boundary/dislocation interactions [6-10]. These twinning related effects need to be considered explicitly to study the deformation behavior of HCP titanium.

Deformation twinning is a sequential process of creation of an embryo (nucleation), propagation of this embryo as a lamella into the crystal or grain, and subsequent thickening (growth) of the twin lamella. Each of these events is stochastic and depends on several factors including microstructural features, internal stresses, direction and rate of loading, strain level, and temperature [4,11-14]. For example, the work of Bieler et al. [4] in pure titanium using in-situ x-ray diffraction shows that the highest resolved shear stress (RSS) twin variant is activated in a grain surrounded by hard grain orientations; whereas low RSS twin variants are activated in grains surrounded by ‘soft’ neighbor orientations. Understanding of twin variant selection is important because it affects the microstructural evolution, such as the formation of parallel twins and/or twin-twin intersections, and then it controls the overall deformation behavior and texture evolution [15]. Most models rely on the average stress state in the grain and a maximum Schmid factor criterion for twin variant selection. However, twin variant activation depends on local stress fluctuations inside the grain and near grain boundaries, and a systematic study is required to establish the statistical distribution of twin variants in a particular material system. Extensive statistical analysis providing the distribution of twin variants in HCP Mg and Zr have been reported [11-12]. An equivalent study for titanium, however, is not available. In this work we report the statistical distribution of twin variants in HCP commercially pure titanium obtained by employing the recently developed automated twinning analysis software METIS to analyze EBSD images of the deformed microstructure [16-17].

Understanding the deformation mechanisms including twinning activity and twin variant selection, requires a predictive crystal plasticity model. Over several decades, several

experimental and computational modeling efforts have been dedicated to understand deformation mechanisms and their effect on the material response of titanium and its alloys [8-9, 18-29]. Most of the experimental literature on titanium concerns the effect of twinning on mechanical anisotropy and strain hardening [5, 8-9, 24, 30-31], and on failure mechanisms [32-33]. A few studies also focus on the effect of interstitial alloying elements on deformation twinning [34].

In what concerns crystal plasticity modeling of Ti, some relevant studies have been performed to understand the deformation mechanisms. The crystal plasticity finite element (CP-FE) approach has been extensively used to study the deformation mechanisms and associated behavior of HCP titanium [9, 24, 26, 28, 31]. The CP-FE studies connect the observed strain hardening behavior and texture evolution to twin and slip activities, and most use an empirical hardening equation like the Voce law. Recent work of Zecevic et al. [29] analyzes the deformation mechanism using CP-FE coupled with a dislocation density based hardening law. The few studies based on the effective medium visco-plastic self-consistent (VPSC) model rely on an empirical hardening law [24, 26, 35]. All of these CP-FE and VPSC works treat twinning in Ti as a deterministic process, and neglect the experimentally observed statistical variations. A stochastic twinning model has been previously developed for HCP Mg and Zr [36-38] that probabilistically accounts for variation in the microstructural and mechanical responses in order to capture the experimentally observed statistical variations in twin nucleation and growth. Successful application of such a stochastic twinning model for titanium requires knowledge of the distribution of microstructural features such as grain size, grain boundary facet area, number of grain boundaries; and the local stress fluctuations due to inter-grain interactions and pre-existing twins [38-42]. All of these distributions need to be incorporated in probabilistic manner leading to several model parameters. To avoid this cumbersome undertaking, in this work we propose a simple Monte Carlo based twin variant selection model to capture the experimentally observed twinning statistics.

In this work, we perform a systematic experimental and computational modeling to understand the deformation mechanisms in rolled commercially pure titanium with a split-basal initial texture. Compression and tension tests are performed along rolling, normal and transverse directions to activate tensile and compressive twinning modes. An automated EBSD twinning analysis software (METIS) is used to analyze and establish the statistical twin variant distributions [16-17]. On the modeling side we develop a simple Monte Carlo twin variant selection scheme, and implement it within the framework of VPSC. Model

parameters are calibrated by fitting the experimentally measured stress-strain response, texture evolution and twin volume fraction evolution. The agreement between observed and predicted twin variant distribution validates the proposed Monte Carlo twin variant selection scheme.

2. Experiments

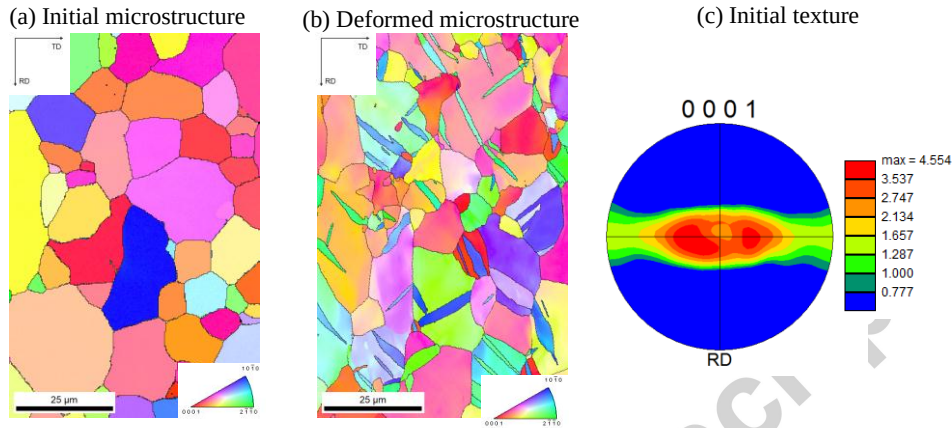


Figure 1. (a) Initial and (b) 10% RD compressed microstructure, and (c) initial texture of commercially pure titanium considered for the present study. Compression along rolling direction to 10% activates deformation twinning. The microstructural images are obtained from the rolling plane and the inverse pole figure color maps corresponds to sample ND direction.

Table 1. Chemical composition of titanium grade 2

Element	Nitrogen	Carbon	Hydrogen	Iron	Oxygen	Titanium
Max. content (%)	0.03	0.06	0.015	0.3	0.25	balance

In this work, rolled commercially pure titanium (grade 2) was used and its chemical composition is given in Table 1. The initial microstructure and texture of the material is shown in Figs. 1(a) and (c), respectively. The average grain diameter is 10.5 μm ($\sim 87 \mu\text{m}^2$ area) with a standard deviation of 6.9 μm, and the initial texture exhibits split-basal poles. To study the anisotropic behavior of HCP titanium in connection with crystal symmetry and deformation mechanisms we performed compression tests along the rolling (RDC), transverse (TDC) and normal (NDC) directions, and tensile test along the rolling (RDT) and transverse (TDT) directions. The gauge dimension of the tensile samples is 12.5x5x5 mm³ and the compression samples is 12x8x8 mm³ with the long dimension corresponds to the loading direction. Tensile and compression tests were performed at a strain rate of 10⁻³ and 10⁻⁴ /s, respectively, at room temperature. For all five tests, different samples were strained to

5%, 10% and 20% for characterization purpose. In-situ or interrupted in-situ methods are suitable to study the evolution of microstructure, twinning, and texture, at a micron scale. However, in this work we are concerned with implementing a numerically efficient approach for simulating texture, mechanical response, and microstructural evolution at a macroscopic scale. We use, for such purpose, different samples that are deformed to different strain levels for each loading condition. Electron Back Scatter Diffraction (EBSD) was used to capture the deformation microstructure and texture. To perform EBSD imaging, deformed samples were mechanically ground with silicon carbide paper to grade 4000 and then electro-polished in the Struers A3 solution. EBSD scans of a $3.0 \times 3.0 \text{ mm}^2$ area (from nine different scans) using a $10 \mu\text{m}$ step size was collected and analyzed using TSL-OIM software to obtain crystallographic textures. Twinning statistics (see Sec. 5 for more detail) were obtained based on $0.2 \mu\text{m}$ step size EBSD scans over total areas of $\sim 0.42 \text{ mm}^2$ (from six different scans of $350 \times 200 \mu\text{m}$ areas each) for each sample. A recently developed automated twin statistics code was employed to analyze the EBSD data [16-17]. In the present study, only {10-12} tensile and {11-22} compression twins were observed, and those are the ones accounted for in the simulations.

Table 2. Volume fraction (in %) of tensile and compression twin determined from EBSD maps and VPSC model (in brackets) at strain of 0.1.

	RDT	TDT	RDC	TDC	NDC
Tensile twins	1.28 (0.4)	6.58 (10.6)	28.68 (19.9)	3.76 (8.2)	5.21 (5.5)
Compression twins	0.99 (0.5)	0.63 (0.3)	0.30 (0.8)	4.16 (2.9)	10.85 (9.8)

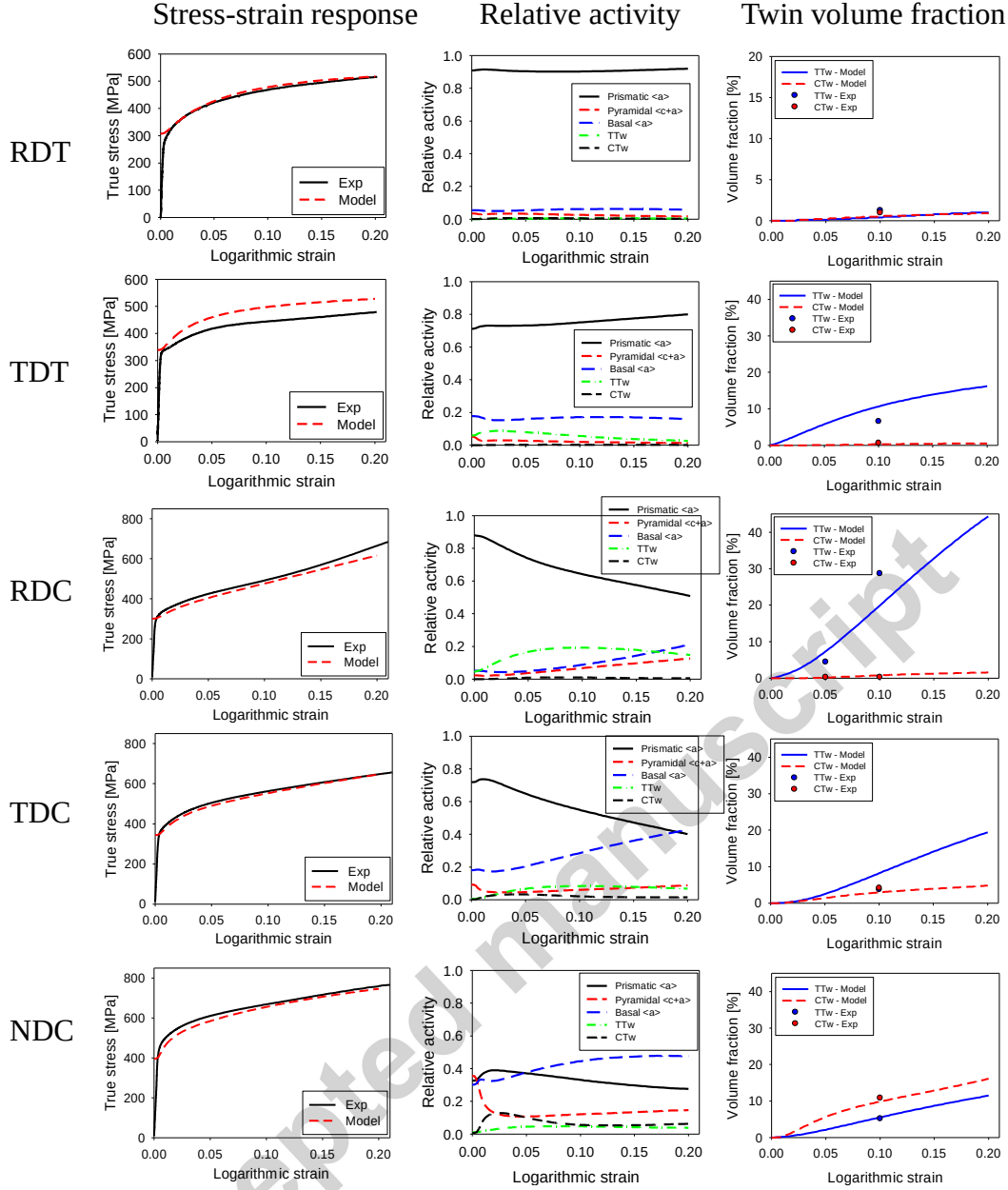


Figure 2. Deformation behavior of polycrystalline titanium under different loading conditions. First column: measured and model predicted stress-strain response; second column: model predicted relative activities of slip and twin modes; third column: measured and model predicted twin volume fraction evolution.

The experimentally measured stress-strain response for all five loading conditions is shown in Fig. 2. The difference in mechanical behavior for tension and compression loading along the RD, TD, and ND is significant. The initial yield stress for RDT and TDT is ~307 and ~338 MPa, respectively. For RDC, TDC and NDC is ~300, 343 and 395 MPa, respectively. The spread of about 25% in yield stresses shows that asymmetry at yield is non-negligible and the spread increases with strain hardening. Similar observations are reported by Hama et al. [31], Nixon et al. [26] and Zecevic et al. [29]. Measured twin volume fraction

at 10% strain using EBSD is shown in the third column of Fig. 2 and also presented in Table 2. As mentioned before, $\{10\text{-}12\}$ tensile and $\{11\text{-}22\}$ compression twins are activated, an example of which are the $\{10\text{-}12\}$ tensile twins is shown for RDC 10% in Fig. 1(b). Specifically, $\{10\text{-}12\}$ tensile twins are mainly activated for RDC and TDT, whereas $\{11\text{-}22\}$ compression twins are activated for NDC, although a small tensile twin activity is also observed in NDC. The evolution of deformation texture with strain for all loading cases is shown in Fig. 3. For RDT, texture evolves gradually and there is no clear evidence of twin reorientation, which is in agreement with the measured twin volume fraction. In RDC and TDT, the twinning reorientation of $\sim 90^\circ$ related to $\{10\text{-}12\}$ tensile twinning is clearly evident. In NDC, the activation of $\{11\text{-}22\}$ compression twin leads to the observed grains reorientation of $\sim 60^\circ$. In TDC, the weak texture evolution may be associated with the small activation of both $\{10\text{-}12\}$ tensile and $\{11\text{-}22\}$ compression twins. Comparing all five loading cases, activation of tensile and compression twins is statistically significant only in RDC and NDC cases, respectively. As a consequence, only these two cases will be considered for further statistical analysis in Sec. 5.

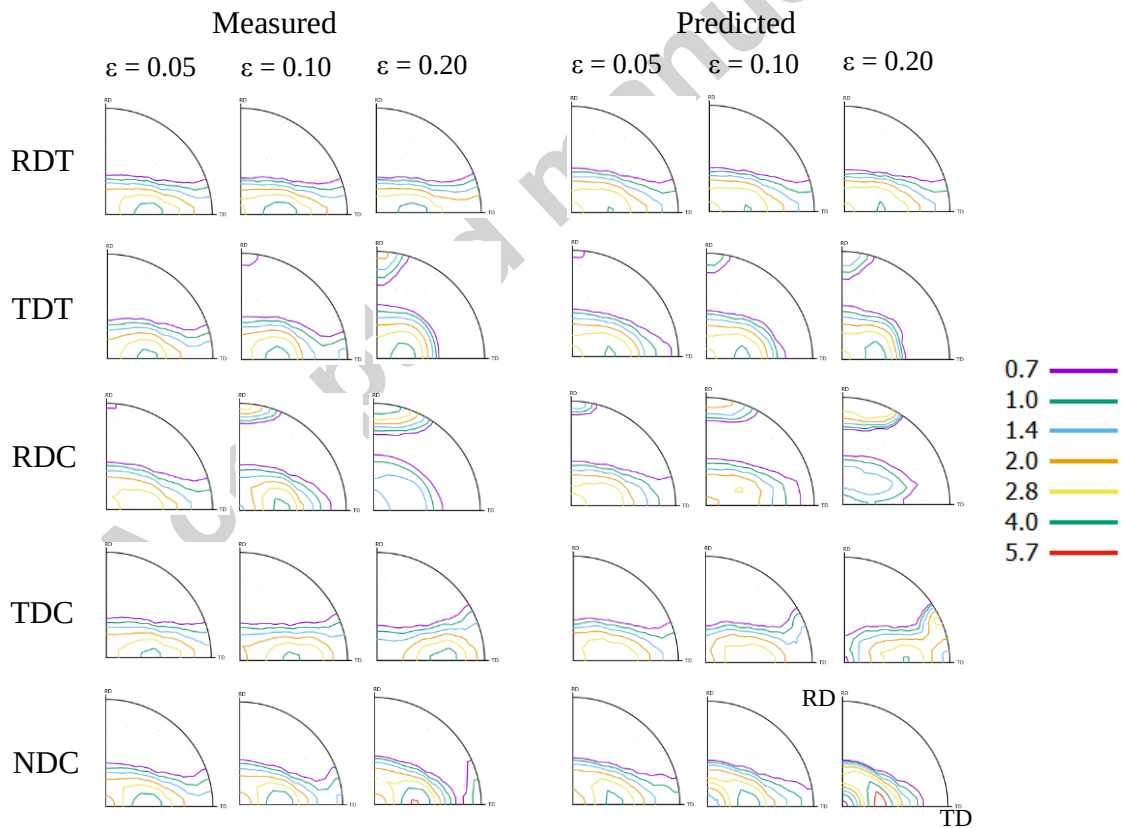


Figure 3. (0001) Pole figures of deformation texture evolution for Ti under different loading conditions as measured by EBSD (left) and predicted by VPSC-DD model (right).

3. Crystal plasticity modeling

To study the underlying active deformation mechanisms of CP-Ti and to provide an insight into the observed twinning and strain hardening behaviors, here we employed a visco-plastic self-consistent (VPSC) model. A comprehensive description of the VPSC model originally developed by Lebensohn and Tome [43] can be found in Lebensohn et al [44]. In Sections 3.1 and 3.2, the polycrystal model and dislocation density based hardening law for slip and twinning systems will be briefly described. The Monte Carlo scheme used for twin variant selection and reorientation is novel and will be described in Section 3.2.

3.1 The Visco-Plastic Self Consistent (VPSC) model

VPSC model describes the polycrystal as a collection of orientations (grains) with associated volume fractions chosen to represent the initial texture of the aggregate. Each grain is regarded as a visco-plastic inclusion embedded in, and interacting with, a “homogeneous effective medium” (HEM), which has the average properties of the aggregate. The macroscopic response of the polycrystal results from the contribution of each grain. The visco-plastic compliance of the HEM is given by a self-consistent condition applied on the grain averages. At the single crystal level, the strain rate of a grain is accommodated by the shear rates of slip and twinning systems, as:

$$\dot{\epsilon}_{ij}^g = \sum_s \mathbf{m}_{ij}^s \dot{\gamma}^s = \dot{\gamma}_0 \sum_s \mathbf{m}_{ij}^s \left(\frac{\mathbf{m}_{kl}^s \cdot \boldsymbol{\sigma}_{kl}^s}{\tau_c^s} \right)^n \quad (1)$$

Here, $\mathbf{m}^s$ and $\dot{\gamma}^s$ is the symmetric Schmid tensor and shear rate of system s . $\dot{\gamma}_0$ denotes a normalized shear rate, n is the inverse strain-rate sensitivity, $\boldsymbol{\sigma}$ is the stress tensor, and τ_c^s is the threshold shear stress or critical resolved shear stress (CRSS) required to activate system s . The latter is given by the constitutive law described in Section 3.2. In order to avoid the further strain-rate dependence associated with the power n in Eq. 1, $\dot{\gamma}_0$ is chosen equal to the norm of the macroscopic strain rate $\|\dot{\epsilon}_{ij}\|$. The constitutive laws relating strain-rate and stress for the aggregate are written in a linearized form as:

$$\dot{\epsilon} = \bar{\mathbf{M}} : \bar{\boldsymbol{\sigma}} + \bar{\dot{\epsilon}}_0 \quad (2)$$

Where $\bar{\mathbf{M}}$ and $\bar{\dot{\epsilon}}_0$ are the macroscopic visco-plastic compliance tensor and back-extrapolated terms for the aggregate, respectively. The inclusion formalism couples stress and strain-rate in the grain ($\boldsymbol{\sigma}^g, \dot{\epsilon}^g$) with the average stress and strain-rate in the effective medium ($\bar{\boldsymbol{\sigma}}, \bar{\dot{\epsilon}}$) through the interaction equation:

$$\dot{\boldsymbol{\varepsilon}}^g - \bar{\boldsymbol{\varepsilon}} = (n^{\text{eff}}(1 - \mathbf{E})^{-1} : \mathbf{E} : \bar{\mathbf{M}}^{\text{secant}}) : (\boldsymbol{\sigma}^g - \bar{\boldsymbol{\sigma}}) \quad (3)$$

$\mathbf{E}$ is the visco-plastic Eshelby tensor, $\bar{\mathbf{M}}^{\text{secant}}$ is the macroscopic visco-plastic compliance tensor for the secant interaction ($\bar{\boldsymbol{\varepsilon}}_0 = \mathbf{0}$), and the parameter n^{eff} “tunes” the stiffness of the inclusion-matrix interaction: $n^{\text{eff}} = 0$ for a Taylor case, $n^{\text{eff}} = 1$ for the stiff secant case, and $n^{\text{eff}} = n$ ($=20$ in our calculations) for the compliant tangent case. In this work, following the systematic analysis of Lebensohn et al. [44], we adopt an intermediate compliance $n^{\text{eff}} = 10$ in Eq. 3.

3.2 Constitutive hardening law

Here we use a thermally activated dislocation density based hardening law (DD model) developed by Beyerlein and Tome [45]. In this model, evolution of threshold shear stress follows the changes in the dislocation density on individual slip systems. Within this DD model, the threshold shear stress for slip τ_c^s is expressed as,

$$\tau_c^s = \tau_{0,f}^s + \tau_{\text{for}}^s + \tau_{\text{deb}}^s \quad (4)$$

where $\tau_{0,f}^s$, τ_{for}^s and τ_{deb}^s is the friction stress (depend on the Peierls stress, precipitates and the initial dislocation content), forest dislocation interaction stress and dislocation substructure interaction stress, respectively. In this work, experiments are performed at a fixed strain rate and temperature, and so we assume constant value for friction stress $\tau_{0,f}^s$. The evolution of forest dislocation interaction stress τ_{for}^s follows the traditional Taylor law [46], as,

$$\tau_{\text{for}}^s = \chi^s b^s \mu^s \sqrt{\rho_{\text{for}}^s} \quad (5)$$

where $\chi^s = 0.9$ is a dislocation interaction parameter, and b^s , μ^s and ρ_{for}^s is the value of the Burgers vector, effective shear modulus and forest dislocation density of system s . The hardening contribution of dislocations stored within substructures, such as cell walls, related to substructure dislocation density ρ_{sub} according to Madec et al. [47]:

$$\tau_{\text{sub}}^s = k_{\text{sub}} \mu^s b^s \sqrt{\rho_{\text{sub}}} \log\left(\frac{1}{b^s \sqrt{\rho_{\text{sub}}}}\right) \quad (6)$$

Here, $k_{\text{sub}} = 0.086$ is an empirical parameter recovering the Taylor law for low substructure dislocation densities [48]. The evolution of the forest density ρ_{for}^s governed by the generation or storage and the removal of dislocations, via:

$$\frac{\partial \rho_{\text{for}}^s}{\partial \gamma^s} = \frac{\partial \rho_{\text{gen,for}}^s}{\partial \gamma^s} - \frac{\partial \rho_{\text{rem,for}}^s}{\partial \gamma^s} = k_1^s \sqrt{\rho_{\text{for}}^s} - k_2^s(\dot{\varepsilon}, T) \rho_{\text{for}}^s \quad (7)$$

where k_1^s is a coefficient for dislocation storage via statistical trapping of gliding dislocations by forest obstacles and k_2^s is a coefficient for dynamic recovery by thermally activated mechanisms, and it is given as [45],

$$\frac{k_2^s}{k_1^s} = \frac{\chi b^s}{g^s} \left(1 - \frac{kT}{D^s b^3} \ln \left(\frac{\dot{\epsilon}}{\dot{\epsilon}_0} \right) \right) \quad (8)$$

Here, k , $\dot{\epsilon}_0$, g^s and D^s are the Boltzmann constant, a reference strain rate, an effective activation enthalpy and a drag stress, respectively. The change in substructure dislocation density is controlled by the rate of recovery of all active dislocations through:

$$\Delta \rho_{\text{sub}} = \sum_s q^s b^s \frac{\partial \rho_{\text{rem,for}}^s}{\partial \gamma^s} |\Delta \gamma^s| \quad (9)$$

where q^s is a dislocation recovery rate that quantifies the fraction of dislocations stored as substructure.

Twinning: Here, we assume that twinning has associated, as slip, a critical resolved shear stress of activation in the twinning plane and along the twinning direction (see Eq. 1). However, twinning differs from slip in its directionality, which we model by allowing activation only if the resolved shear stress is positive (along the shear direction of the twin). Similar to slip, the evolution of threshold shear stress or CRSS of twinning τ_c^t is a sum of a temperature- and rate-independent frictional stress and a rate-independent latent hardening term that coupling slip and twin systems as,

$$\tau_c^t = \tau_0^t + \mu^t \sum_t C^{st} b^t b^s \rho_{\text{for}}^s \quad (10)$$

3.3 Variant selection and twin reorientation model

An aspect of twinning that needs to be incorporated into the models is the fact that twins are domains inside the grain (usually of lamellar morphology) with a different orientation than the surrounding matrix. These twinned regions contribute to the texture of the aggregate and also act as effective barriers for the propagation of dislocations or other twin lamellae. Based on the practical limitation that simulations cannot treat every twin as a new orientation, in the past we have implemented a Predominant Twin Reorientation (PTR) scheme [49] in the VPSC framework to account for twinning reorientation. The PTR criterion consists on finding the most active twin system in the grain, and reorienting the whole grain according to such system, provided other conditions are also met. In this work we introduce a variation of the PTR model based on a Monte-Carlo (MC) selection criterion of the active twin variant which, while favoring the predominant twin system, may also predict reorientation on less favorable twin variants.

As in the case of the PTR scheme, within each grain g we keep track of the shear strain $\gamma^{t,g}$ contributed by each twin system t , and of the associated volume fraction $V^{t,g} = \gamma^{t,g}/S^t$ (S^t is the characteristic twin shear). The sum over all twin systems associated with a given twin mode, and over all grains, represents the 'accumulated twin fraction' $V^{acc,mode}$ in the aggregate for the particular twin mode (the one that would be measured by SEM).

$$V^{acc,mode} = \sum_g \sum_t \frac{\gamma^{t,g}}{S^t} \quad (11)$$

At each incremental step, some grains are fully reoriented by twinning provided certain conditions are fulfilled. We call 'effective twinned fraction' $V^{eff,mode}$ the volume associated with the fully reoriented grains for that mode, and define a threshold volume fraction $V^{th,mode}$

$$V^{th,mode} = A^{th1} + A^{th2} \frac{V^{eff,mode}}{V^{acc,mode}} \quad (12)$$

which has to be exceeded for grains to be reoriented. The algorithm given by Eq. 12 prevents grain reorientation by twinning until a minimum volume fraction of twinning, A^{th1} , is accumulated in any given system. In the calculations presented here we use a rather low value of the threshold $A^{th1}=0.02$ (typically $A^{th1}=0.10$ in the PTR calculations), and a value $A^{th2}=0.50$ for the upper threshold. As twin shear takes place $V^{acc,mode}$ increases and the threshold $V^{thres, mode}$ decreases. After each deformation increment we pick grains at random, and identify in each the twin system with the highest accumulated volume fraction (PTS). If the latter is larger than the threshold $V^{thres, mode}$ then the grain is allowed to reorient, and $V^{eff, mode}$ and $V^{thres, mode}$ are updated. As more grains are fully reoriented by twinning, $V^{eff, mode}$ increases and so does the threshold. The process is repeated until either all grains are randomly checked or until the effective twin volume exceeds the accumulated twin volume. In the latter case we stop reorientation by twinning and proceed to the next deformation step. A characteristic of this process is that the twinned fraction is consistent with the shear activity that the twins contribute to deformation. Note that the predicted twin reorientation, and so the evolution of twin fraction, depends very weakly on the choice of parameters A^{th1} and A^{th2} . The reason is that the PTR and the MC schemes are predominantly controlled by the feedback represented by the condition that $V^{eff,mode}/V^{acc,mode}$ should not exceed the value 1.

Even values of $A^{th1}= 0.0$ and $A^{th2} = 1.0$ are used, there is little difference with the results reported in Figs. 2 and 3.

In the PTR scheme, only the historically most active twin system in each grain is considered for reorienting the whole grain by twinning. In the MC scheme every, twin system (variant) has a chance, proportional to the accumulated volume fraction that each would have associated. The computational procedure consists of ‘aligning’ the twin fraction of each system in sequential intervals proportional to $V_{fr}^{tw1}, V_{fr}^{tw2}, \dots, V_{fr}^{twn}$, normalizing them with the sum, picking a random number $0 < r < 1$, and choosing the twin whose segment contains r . This procedure gives a larger probability to the predominant twin in the grain (first variant), but also assigns non-negligible probability to lesser variants. A result of this procedure is illustrated in Fig. 4 for the case of RDC after 20% strain, to be discussed later in Section 4.4.

Observe that while PTR always picks the most favorable variant, twins with lesser Schmid factors are selected by the MC scheme. We will show later that this approach predicts distributions of twin variants that closely resemble the experimentally observed distributions. The result can be interpreted as an empirical way to capture the role-played by stress fluctuations inside grains in activating twinning. The MC scheme is an efficient, although empirical, way to introduce fluctuations not addressed by the effective medium approach in which VPSC is based. Niezgoda et al. [38] and Arul Kumar et al. [36] have proposed methods for introducing stress fluctuations at grain boundaries and at grain interiors, respectively, based on computationally complex preliminary calculations of distributions.

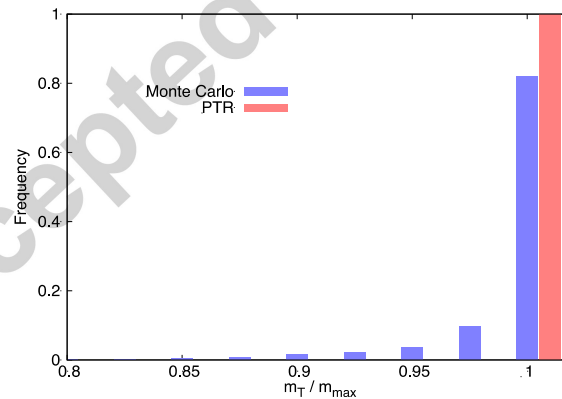


Figure 4. Comparison between PTR and Monte Carlo schemes for tensile twin variant selection associated with RDC. Frequency distribution of relative Schmid factor (defined as: Schmid factor of twin used for reorienting grain / maximum Schmid factor among all twin variants). The result shows the activation of lower Schmid twin variants in the Monte Carlo scheme, in contrast with the PTR scheme.

4. Model predictions

A visco-plastic self-consistent (VPSC) model, coupled with a Monte Carlo (MC) twin variant selection scheme, is employed here to predict the deformation behavior of polycrystal

titanium. The initial discrete texture consists of 15552 grains that reproduce the split-basal texture shown in Fig. 1(c). In the model, plastic deformation is accommodated by prismatic $\langle a \rangle$: $\{10\bar{1}0\}\{11\bar{2}0\}$, basal $\langle a \rangle$: $\{0001\}\{11\bar{2}0\}$ and pyramidal $\langle c + a \rangle$: $\{10\bar{1}1\}\{11\bar{2}3\}$ slip modes, and tensile: $\{10\bar{1}2\}\{10\bar{1}1\}$ and compression: $\{11\bar{2}2\}\{11\bar{2}3\}$ twins. Parameters of the dislocation density based hardening law presented in Sec. 3.2 are obtained by fitting the experimentally measured stress-strain response, texture evolution and twin volume fraction evolutions of all five loading cases (RDC, TDC, NDC, RDT and TDT) together. However the twinning statistics presented in Sec.5 is not used in the fitting process. The constitutive law parameters are listed in Tables 3 and 4 for slip and twinning modes, respectively.

Table 3. Hardening model parameters controlling the evolution of critical shear stresses of slip systems.

	$\alpha=1$ Prismatic $\langle a \rangle$	$\alpha=2$ Basal $\langle a \rangle$	$\alpha=3$ Pyramidal $\langle c+a \rangle$
b^α [m]	2.9511×10^{-10}	2.9511×10^{-10}	5.5364×10^{-10}
τ_0^α [MPa]	90.5	170	210
k_1^α [m ⁻¹]	2.5×10^8	1.5×10^9	5.0×10^9
g^α	0.002	0.002	0.0055
D^α [MPa]	100	100	100

Table 4. Hardening model parameters controlling the evolution of critical shear stresses of twin systems.

	$\beta=1$ Tensile twin	$\beta=2$ Compression twin
b^β [m]	6.0040×10^{-11}	8.2015×10^{-11}
τ_0^β [MPa]	180	250
τ_{prop}^β [MPa]	180	250
$C^{1\beta}$	100	1000
$C^{2\beta}$	50	50
$C^{3\beta}$	250	170

The stress-strain response, relative activities of deformation modes and the evolution of twin volume fraction predicted by the VPSC-MC model for all five loading cases is shown in the first, second and third columns of Fig. 2, respectively. Except for TDT, the predicted stress-strain response shows good agreement with the experimental measurement. The over-prediction of the TDT flow stress may be due to incipient necking in the experimental test, which the polycrystal model would not reproduce. If it were coming from deformation localization close to grain boundaries, local FE modeling of the same system reported

recently by Zecevic et al. [29] should capture it. However, the results of Zecevic et al. [29] also show a tendency to over-predict TDT, and clearly over-predict NDT. In RDT, prismatic $\langle a \rangle$ slip mainly accommodates the plastic deformation with small contribution from basal $\langle a \rangle$ and pyramidal $\langle c+a \rangle$ slip, and twinning activity is practically negligible over the entire deformation range, which is in good agreement with the experimental observation (see Table 2). TDT is also dominated by prismatic $\langle a \rangle$ slip with the addition of tensile twinning activity. The model predicts a tensile twin volume fraction of 10.6% at 10% strain, which is higher than the measurement (6.6%). In RDC, the early stages of deformation are mainly accommodated by prismatic slip and tensile twinning with both basal and pyramidal slip modes being activated at higher strains. Here the evolution of tensile twin volume fraction increases steadily with strain. The predicted tensile twin volume fraction is $\sim 29\%$ at 10% strain, which is higher than the measurement (20%). In TDC, the plastic deformation is accommodated by prismatic and basal slip modes with a small contribution from twinning. Similar to TDT and RDC, the model predicts slightly higher twin volume fractions compared to the experimental observations. For NDC, all three-slip modes are activated at low strains and the contribution of pyramidal slip decreases with increasing compression twin activity. Compression twinning is the more active twinning mode for NDC with a lesser contribution from tensile twinning. Model predicted volume fractions of tensile and compression twins at 10% strain agree with the experimental observations.

The model predicted deformation texture evolution for all five loading cases at strain levels of 5%, 10% and 20% as shown in Fig. 3 along with EBSD based experimental textures. As discussed in Sec. 2, the key features of the texture evolution are connected with twinning activities. A comparison between model predictions and experimental measurement shows, in general, very good agreement. Predicted deformation mechanisms in the present work are qualitatively similar to the recent work of Zecevic et al. [29] that used an elasto-plastic self-consistent scheme with a dislocation hardening law similar to ours, implemented in a finite element framework.

5. Twin variant statistics

In this section, we analyze the twin variant statistics in titanium extracted from the EBSD images, as mentioned in Sec. 2. A recently developed automated twinning analysis tool is employed to analyze the microstructure [16-17]. Developing reliable statistical

correlations associated with twinning behavior requires analysis of a large number of twins and grains. As a consequence, only the RDC and NDC cases are considered for $\{10-12\}$ tensile and $\{11-22\}$ compression twins, respectively, at a strain of 10%. The 0.42 mm^2 area (5 scans each) with 0.2 micron steps on the RDC and NDC samples gave a total of 4678 grains and 2128 tensile twins for RDC and 4249 grains and 1198 compression twins for NDC. A separate and in-depth analysis of twinning, variants and twin-twin junctions for this system can be found in [49].

For tensile and compressive twins, all six variants are considered in the analysis, and the twin variants in each grain ($v_i, i = 1,6$) are classified in order of decreasing Schmid factor. In the EBSD twinning analysis, the Schmid factor for each twin variant is calculated using the macroscopic tensile or compressive stress direction, and the parent grain orientation. Accordingly, $v_{(1)}$ and $v_{(6)}$ are the highest and lowest Schmid factor variants, respectively. The EBSD-extracted twin variant frequency distribution of tensile twins for RDC and compression twins for NDC is shown in the left side of Fig. 5. Tensile twin variants 1 and 2 seem to be activated in the majority of grains ($\sim 77.2\%$), along with small contributions from the other variants. In addition, variant 1 (54.1%) is highly active compared to variant 2 (23.1%). On the other hand, the first three variants of compression twins are equally active in NDC ($v_{(1)} = 29.6\%$, $v_{(2)} = 30.9\%$ and $v_{(3)} = 17.6\%$) and, similar to tensile twins, the last three variants are statistically less active. These results clearly provide evidence for the nucleation of low variant twins in the polycrystal, and the trend is stronger for the compression twins compared to tensile twins. The distribution of area fraction for all variants of tensile and compressive twins is shown in the right column of Fig. 5. Similar to the frequency distribution, the first two variants of tensile twins, and the first three variants of compression twin have high twin volume fractions compared to other low variants.

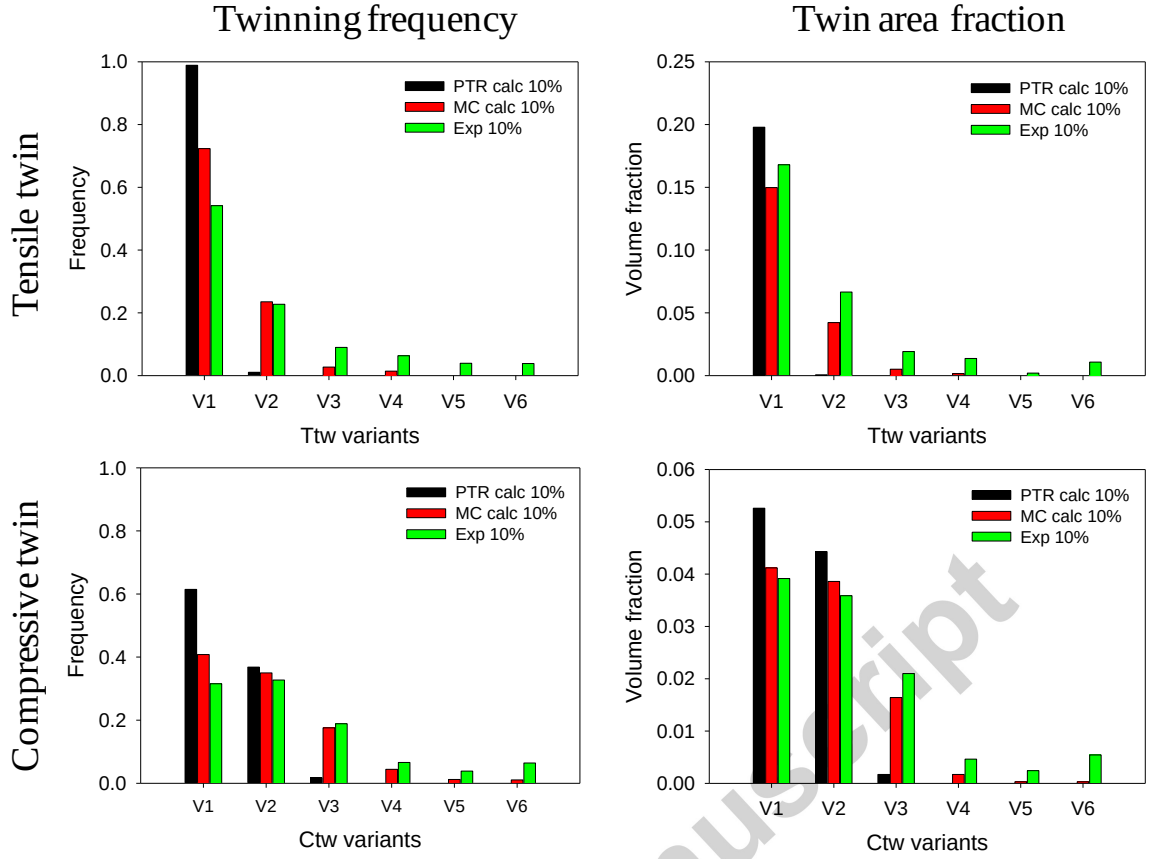


Figure 5. Frequency and area fraction statistics of tensile and compression twins obtained from RDC and NDC loading conditions, at a strain of 10% compression show predominance of the first two most highly favored variants, with larger observed experimental contributions of the other four variants than predicted by the model.

The experimental and model predicted twinning frequencies and twin volume fractions for both, tensile and compression twins are compared in Figure 5. In addition to the VPSC-MC model predictions, VPSC-PTR predicted distributions are also shown in Fig. 5 to illustrate the effectiveness of the simple Monte Carlo scheme. In the VPSC model (both PTR and MC scheme), twin variants are identified using the individual grain stress, which is different than the macroscopic stress (see Sec. 3.3), while experimentally observed twin variants are classified using the macroscopic loading condition. To directly compare the model predictions with the experiment, we have re-ranked the model selected twin variants using the macroscopic polycrystal stress instead of individual grain stresses. The PTR scheme primarily activates variant 1 tensile twins and variant 1 and 2 in compression twins, which is different from the experimental observations. Note that the PTR scheme will always activate variant 1 with respect to an individual grain stress, and the activation of variant 2 seen in the compression twins in Fig. 5 is an effect of re-ranking of twin variants using the polycrystal

average stress. Compared to the PTR scheme, the Monte Carlo scheme activates more low variant twins in agreement with the experimental observations, but to lesser extent than experimentally observed. For the twin volume fraction distributions, the VPSC-MC model predictions are also in better agreement with the experiments compared to the VPSC-PTR model. Although the proposed stochastic scheme is simple, it captures the experimentally observed lower Schmid factor twin variant activation.

6. Discussion

Implications of MC scheme: Several studies have been dedicated to understanding the twin variant selection process in HCP Ti [28, 51-54] using different criteria such as minimization of deformation energy, neighboring grain slip and/or twin induced twin nucleation, and neighboring grain accommodation of twinning shear. The most common conclusion in all of these studies is that local stress fluctuations and the microstructural heterogeneity in the polycrystal cause the activation of low Schmid factor twin variants for both tensile and compression twin modes. Several local processes can develop stress fluctuations in a grain such as i) grain-grain interactions; ii) neighboring grain twins induced local stresses; iii) non-uniform stresses induced by dislocation interactions. The VPSC model cannot realize such local intra-granular mechanisms, and so they have to be incorporated into the model in an ad-hoc fashion. As mentioned in Sec. 1, a complex twin stochastic model was developed to explicitly account for the local stress fluctuations and microstructural heterogeneity [36-38]. But in this work, we have introduced a stochasticity associated with twinning via a simple Monte Carlo scheme for twin variant selection. The arbitrary random selection used to identify the twin variant indirectly represents the twinning probability associated with local stress fluctuations and microstructural heterogeneities. Even if the proposed MC scheme simplifies the complex twinning process into a simple random number selection, it succeeds qualitatively in activating the statistically correct twin variants for both tensile and compression twin modes as seen in Fig. 5.

Low SF variants are more likely for compression twin mode than tensile twin mode: The detailed EBSD based statistical analysis and the model predictions of twin variant distribution shown in Fig. 5 suggest that more low variant twins are activated for {11-22} compression twins compared to {10-12} tensile twins. It can be shown, via a simple Schmid factor analysis, that the reason is the relative inclination of twin variants with respect to loading. For

this analysis, we used the same 15552 grains polycrystal from the simulations (see Fig. 1b). The Schmid factor for each of the six {10-12} tensile and {11-22} compression twins was calculated for compression along the rolling and normal directions, respectively. The twin variants with negative Schmid factors are removed from the analysis. To understand the activation of lower variants, we can plot the distribution of those variants' SF versus the SF of variant-1. The distribution of the SF of variant-3 of tensile twins is shown in the insert of Fig. 6 as a function of the SF of variant-1. If the SF of variants 2 to 6 is close to the maximum SF, then chances are favorable that local stress fluctuations may lead to their activation. We quantify the chance of activating a particular variant $v_{(i=2...6)}$ instead of or together with variant-1 using the following equation.

$$f_i = \frac{\sum_{g=1}^N \begin{cases} 1 & \text{if } m_i^g > 0.15 m_1^g \\ 0 & \text{Otherwise} \end{cases}}{N}, \quad i = 2...6 \quad (13)$$

Where m_i^g and m_1^g are the Schmid factor of variant- i and variant-1 of grain g , and N is the total number of grains. The expression is calculated for all variants of tensile and compression twins separately, and the choice of factor 0.15 in the above equation is arbitrary. The calculated chances of activating twin variants $v_{(i=2...6)}$ instead of or together with variant-1 is shown in Fig. 6. It clearly suggests that the chances for the activation of lower SF twin variants is high for compression twins during NDC compared to tensile twinning during RDC. This is due to the orientation of the twinning systems relative to the axial loading conditions considered in this study. Tensile twins are at about 45° with respect to the c -axis, and can be activated either by applying tension along c -axis or compression perpendicular to the c -axis. Tension along the c -axis gives the same SF for all six variants, but compression perpendicular to the prismatic plane will give similar high SFs for only two co-zonal twin variants. In the case of compression twins, the angle between the twin plane and the c -axis is about 30° . Compression along the c -axis can activate all six variants, but tension perpendicular to the c -axis will tend to induce larger SF in non-co-zonal twins. For the current texture in the NDC case, most of the grain c -axes are subjected to compression with a slight tilt due to spread in the basal texture, and can activate more than two variants based on SF analysis. In the RDC case, most of the grains are subjected to prismatic plane compression and so it can activate only two high SF variants.

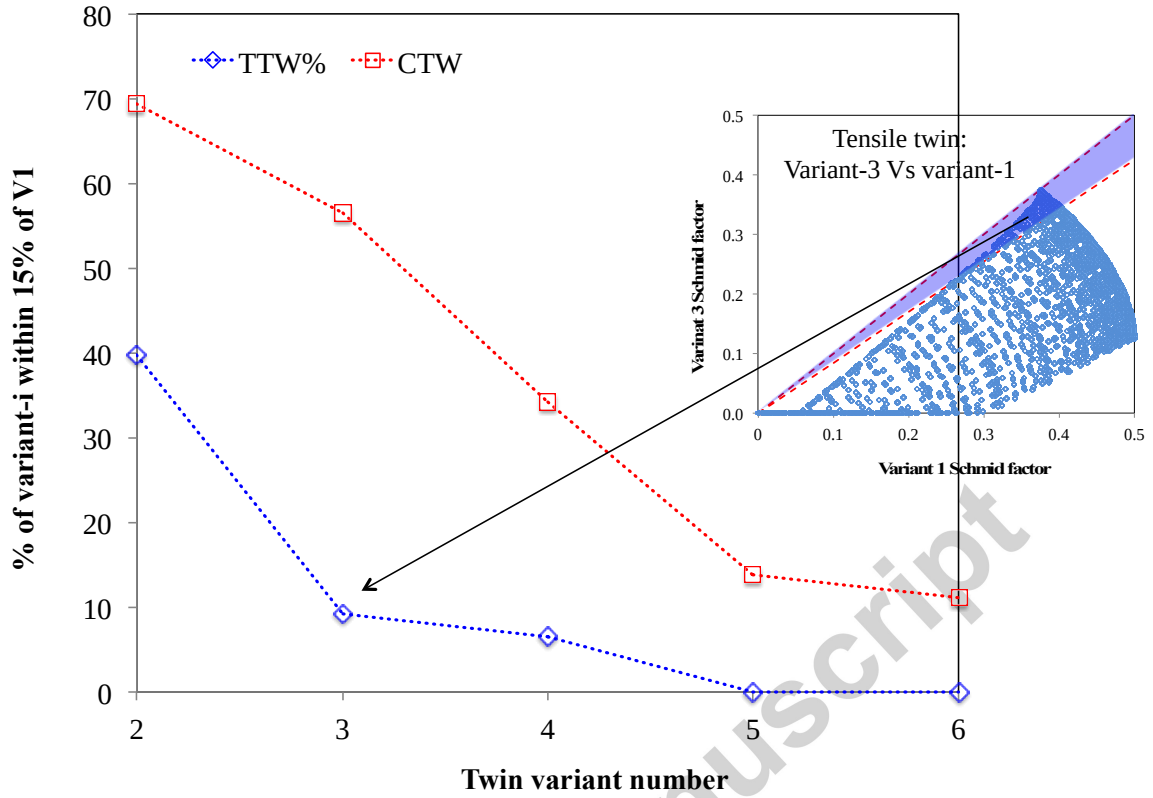


Figure 6. Percent probability of activating variants $v_{(i=2...6)}$ instead of or together with variant-1 for tensile and compression twin modes. The insert shows the distribution of variant-3 SF as a function of variant-1 SF for tensile twins. The overlap between the population (blue) and the shaded domain shows several grain orientations where the v_3 SF is within the 15% band of v_1 .

7. Summary

The deformation behavior of HCP CP-titanium was studied using experimental and computational modeling, with an emphasis on understanding twin variant activation and statistics. Tensile and compression tests along different directions relative to a split basal texture were performed to study the stress-strain response and microstructure and texture evolution. EBSD based statistical of twinning in the deformed microstructures was performed to establish the distribution of twin variants. The VPSC model was used to simulate the deformation behavior, and a simple Monte Carlo twin variant selection scheme was proposed to account for the stochasticity associated with twinning. The proposed VPSC-MC model captures the experimentally measured stress-strain response, texture evolution, twin volume fraction evolution and twin variants statistics.

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